Lecture 18: Parametrizations of Plane Curves

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

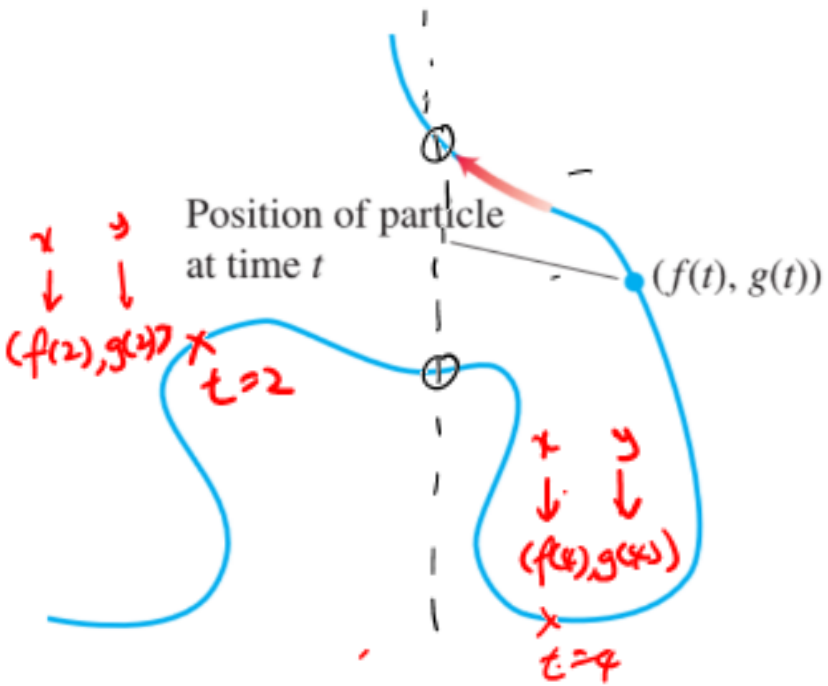
- ☐ Understand the concept of a curve defined parametrically.
- ☐ Convert between parametric and Cartesian equations when possible.
- ☐ Sketch curves given by parametric equations and describe their orientation.
- ☐ Analyze the behavior of a parametric curve over a given interval of the parameter.
- ☐ Identify and graph parametric representations of common and piecewise curves.

Parametric Equations

So far, what we learned is $f: x \to y$

We sometimes define relation between x and y in terms of a third party called a parameter.

Instead of describing a curve as the graph of a function, we think of it as a path of a moving particle whose position is changing over time.



- The curve fails the vertical test.
- we cannot describe it as a function of x.
- Instead, we can describe the curve as $(x, y) = (f(t), g(t))$ for some continuous function f and g.
- $(f(t), g(t))$ would indicate the location of the particle at the time t.

Parametric Equations

If x and y are given as functions

f and g are both continuous function over interval I.

$$x = f(t), \qquad y = g(t)$$

over an interval I of t-values, then the set of points $(x, y) = (f(t), g(t))$ defined by these equations is a parametric curve. The equations are parametric equations for the curve.

t : parameter of the curve

I : parameter interval.

If $I = [a, b]$ (closed interval)

$(f(a), g(a))$: initial point.

$(f(b), g(b))$: terminal point.

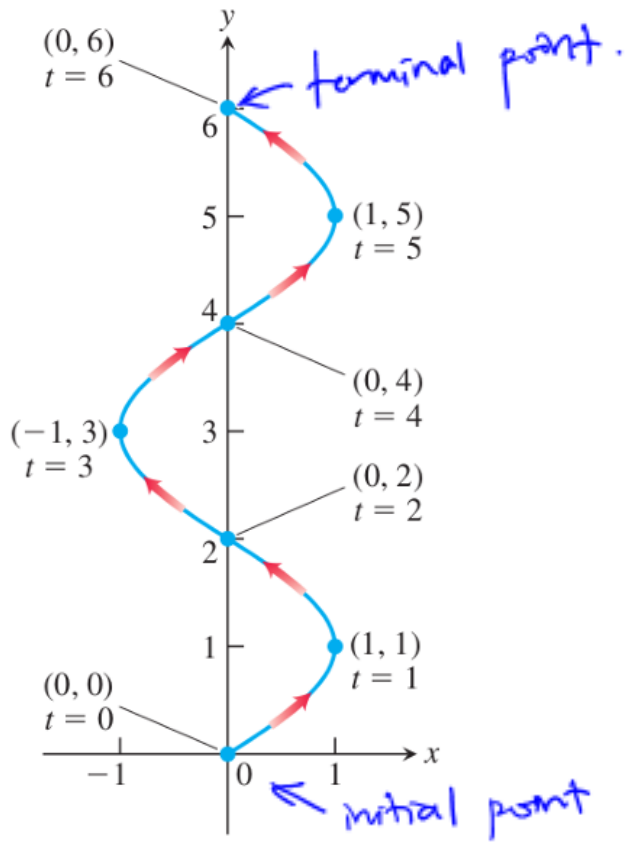
Parametric Equations

Example: Sketch the curve defined by the parametric equations

$$x = \sin\frac{\pi t}{2}, \quad y = t, \quad 0 \le t \le 6.$$

TABLE 10.1 Values of $x = \sin \pi t/2$ and $y = t$ for selected values of t.

t	x	y
0	0	0
1	1	1
2	0	2
3	−1	3
4	0	4
5	1	5
6	0	6



Parametric Equations

Example: Sketch the curve defined by the parametric equations

$$x = t^2, \quad y = t + 1, \quad -\infty < t < \infty.$$

we do not have initial and terminal points here.

$\Rightarrow t = y - 1$

$\Rightarrow x = (y-1)^2$ ← hyperbola.

TABLE 10.2 Values of $x = t^2$ and $y = t + 1$ for selected values of t.

t	x	y
−3	9	−2
−2	4	−1
−1	1	0
0	0	1
1	1	2
2	4	3
3	9	4

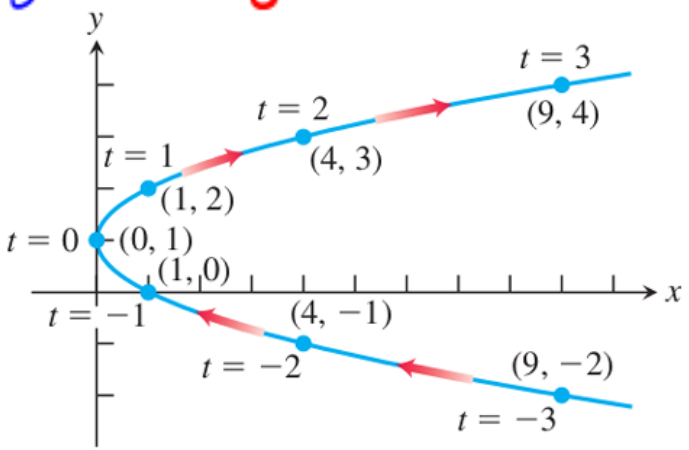


FIGURE 10.3 The curve given by the parametric equations $x = t^2$ and $y = t + 1$ (Example 2).

Parametric Equations

Example: Graph the parametric curves

$\cos^2 t + \sin^2 t = 1$

(a) $x = \cos t, \quad y = \sin t, \quad 0 \le t \le 2\pi.$

(b) $x = a\cos t, \quad y = a\sin t, \quad 0 \le t \le 2\pi.$

(a)

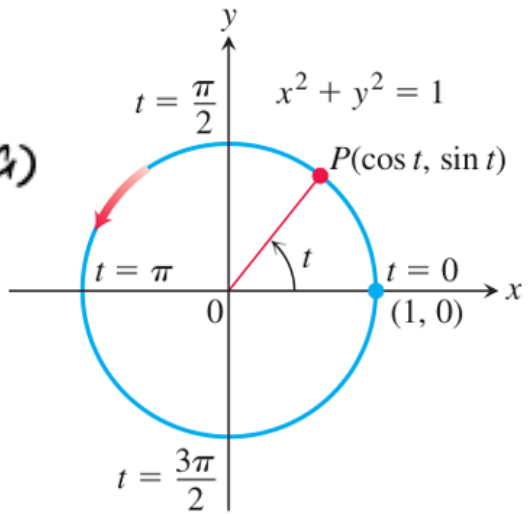


FIGURE 10.4 The equations $x = \cos t$ and $y = \sin t$ describe motion on the circle $x^2 + y^2 = 1$. The arrow shows the direction of increasing t (Example 3).

(a) $x^2 + y^2 = \cos^2 t + \sin^2 t = 1$

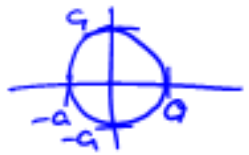
(Circle of radius 1 with the center at origin)

(b) $x^2 + y^2 = (a\cos t)^2 + (a\sin t)^2$

$= a^2(\cos^2 t + \sin^2 t)$

$= a^2$

(circle of radius a with the center at origin



Parametric Equations

Example: The position $P(x, y)$ of a particle moving in the xy-plane is given by the equations and parameter interval

$$x = \sqrt{t}, \quad y = t, \quad t \geq 0.$$

Identify the path traced by the particle and describe the motion.

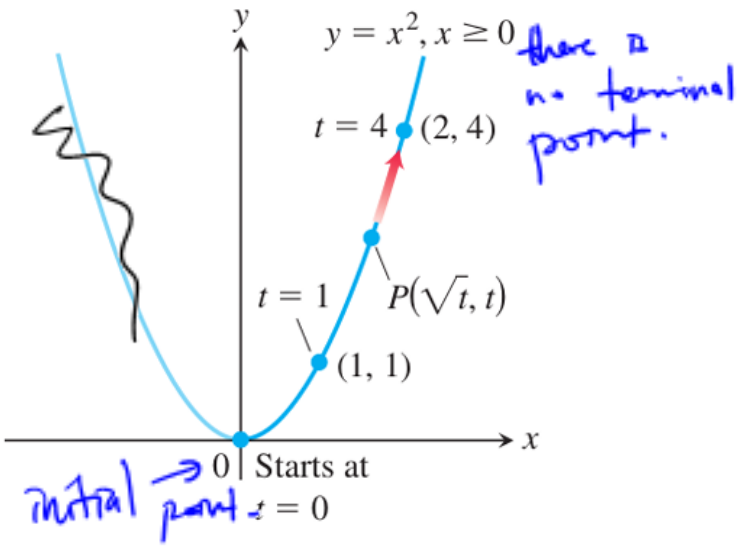


FIGURE 10.5 The equations $x = \sqrt{t}$ and $y = t$ and the interval $t \geq 0$ describe the path of a particle that traces the right-hand half of the parabola $y = x^2$ (Example 4).

$y = t = (\sqrt{t})^2 = x^2$ ← parabola.

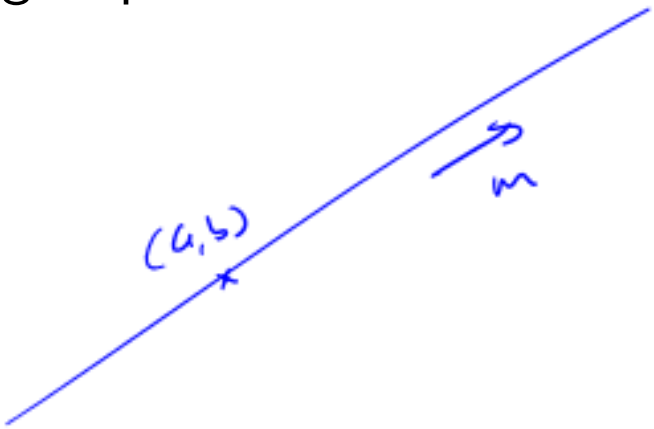
$t \geq 0 \Rightarrow x = \sqrt{t} \geq 0$

The curve involves only the right half of the parabola $y = x^2$.

So, always do not forget to check the interval for t after you identify the curve.

Parametric Equations

Example: Find a parametrization for the straight line through the point (a, b) having slope m.



Equation of a straight line is

$$(y-b) = m(x-a)$$

If we let $t = x-a$, then

$$x = t+a$$

$$y-b = mt \Rightarrow y = mt+b, \quad -\infty < t < \infty$$

(parametric equation)

This is also a parametric equation of the line.

Alternatively, another possible way to do parametrization is

Let $x = t$

$$y-b = m(t-a)$$

$$y = m(t-a)+b, \quad -\infty < t < \infty$$

* Parametrization is not always unique.

Parametric Equations

Example: Sketch and identify the path traced by the point $P(x, y)$ if

$$x = t + \frac{1}{t}, \quad y = t - \frac{1}{t}, \quad t > 0.$$

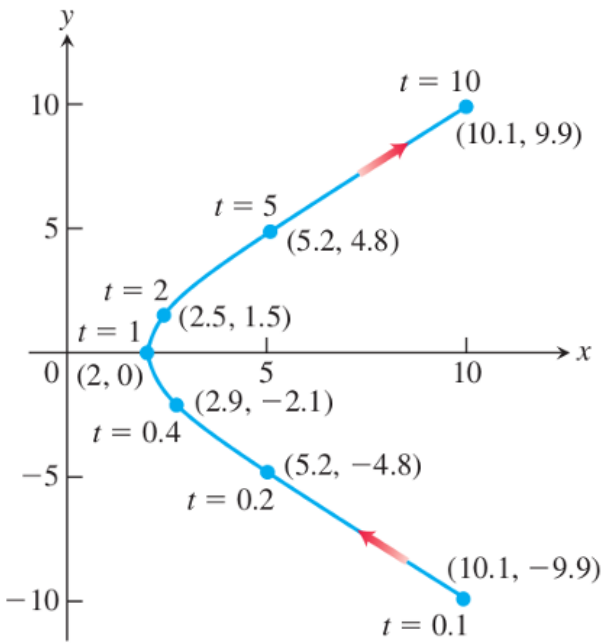


FIGURE 10.7 The curve for $x = t + (1/t)$, $y = t - (1/t)$, $t > 0$ in Example 7. (The part shown is for $0.1 \le t \le 10$.)

It is hard to find the Cartesian equation by direct substitution.

$$x + y = \left(t + \frac{1}{t}\right) + \left(t - \frac{1}{t}\right) = 2t.$$

$$x - y = \left(t + \frac{1}{t}\right) - \left(t - \frac{1}{t}\right) = \frac{2}{t}.$$

$$(x+y)(x-y) = (2t)\left(\frac{2}{t}\right)$$

$$\Rightarrow \quad x^2 - y^2 = 4 \leftarrow \text{hyperbola.}$$

Parametric Equations

Any function can be rewritten parametrically in many different ways

Example: $y = x^3 + 7$. , $x > 0$

(I) Let $x = t$, $y = t^3 + 7$, $t > 0$

(II) Let $x = \sqrt{t}$, $y = t + 7$, $t > 0$

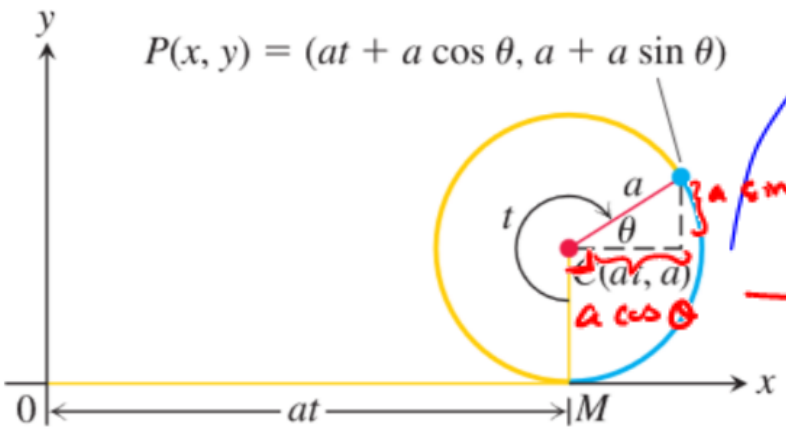
(III) Let $x = e^t$, $y = e^{3t} + 7$, $-\infty < t < \infty$

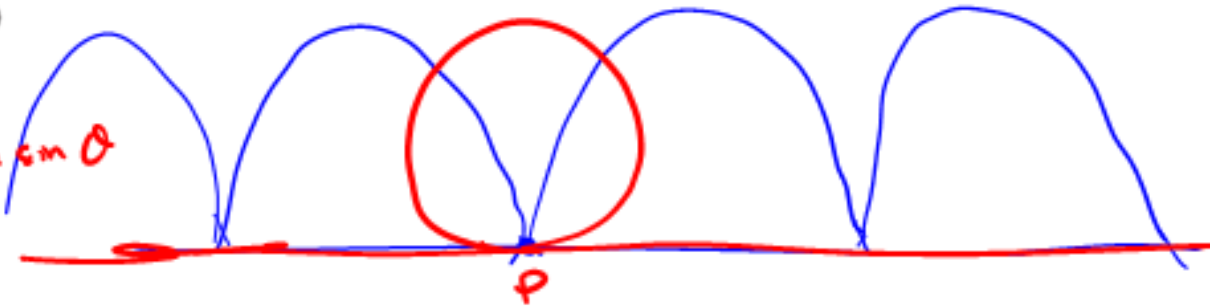
} all of these are continuous function.

and so many more...

Cycloids

A wheel of radius a rolls along a horizontal straight line. Find the parametric equations for the path traced by a point P on the wheel's circumference. The path is called a cycloids.





t: angle through which the wheel turns.

$C(at, a)$: center of the wheel.

Coordinate of P : $(at + a\cos\theta, a + a\sin\theta)$

Observe that $t + \theta = \frac{3\pi}{2} \Rightarrow \theta = \frac{3\pi}{2} - t$.

So $\cos\theta = \cos\left(\frac{3\pi}{2} - t\right) = \cos\frac{3\pi}{2}\cos t + \sin\frac{3\pi}{2}\sin t$.

$= -\sin t$.

$\sin\theta = \sin\left(\frac{3\pi}{2} - t\right) = \sin\frac{3\pi}{2}\cos t - \cos\frac{3\pi}{2}\sin t$.

$= -\cos t$.

So, $x = at + a(-\sin t)$, $y = a + a(-\cos t)$

$$\boxed{x = a(t - \sin t), \quad y = a(1 - \cos t)}$$

parametric equations of the cycloide.